

KalamBot AI - Important Questions for CST302

Ranked by frequency across past year KTU papers.

Module 1: Process Management

[Rank 1 | Asked 12 times]

Explain the different states of a process with a neat diagram.

[Rank 2 | Asked 8 times]

What is a PCB? Explain its components.

[Rank 3 | Asked 5 times]

Differentiate between short-term, medium-term, and long-term schedulers.

Module 2: Process Synchronization

[Rank 1 | Asked 15 times]

Explain the producer-consumer problem and how it is solved using semaphores.

[Rank 2 | Asked 10 times]

What is the critical section problem? Explain the requirements for its solution.

Module 3: Memory Management

[Rank 1 | Asked 14 times]

Explain paging and segmentation with examples.

[Rank 2 | Asked 11 times]

Discuss the different page replacement algorithms (FIFO, LRU, Optimal).

Module 4: Deadlocks

[Rank 1 | Asked 18 times]

Explain Banker's algorithm for deadlock avoidance with an example.

[Rank 2 | Asked 9 times]

What are the necessary conditions for a deadlock to occur?